

Exercise 6 for 'Computational Physics - Material Science', SoSe 2025
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Please provide a well documented submission of your solution. Your submission should include

- A pdf file containing the solution to the questions with the corresponding equations that are implemented in your codes. Figures must contain axis titles with corresponding units and a caption.
- The source code should be commented, and the equations given in the pdf file have to be referenced in the source code.
- There is no need to provide the trajectory files.
- In case your code is not working properly, please provide a description of the debugging attempts you did.

Exercise 6.1: Equation of State

The objective of this exercise is to enrich the NVT engine implemented in the previous exercise with a routine that evaluates the pressure, P , of the system through the internal virial. Once implemented and validated, the equation of state, P versus density ρ , of a LJ fluid is calculated. The *internal* virial, ϑ , is defined as

$$\vartheta \equiv \sum_{i=1}^N \mathbf{r}_i \cdot \mathbf{f}_i = \sum_{i=1}^N \sum_{j>i}^N \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}$$

where $\mathbf{f}_i$ is the total force acting on the i -th atom, $\mathbf{f}_{ij} = -\frac{dU(r_{ij})}{d\mathbf{r}_{ij}}$ is the pair force acting between atoms i and j because of the pair potential, U , and $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. Either of the expressions above can be used to calculate ϑ from simulations. With periodic boundary conditions (PBC), the following should be carefully considered:

- $\mathbf{r}_i$ is the *PBC-wrapped* (that is, minimum image) coordinate of particle i .
- $\mathbf{r}_{ij}$ is the *PBC-wrapped* distance vector between particles i and j .

From the virial theorem in the context of classical statistical mechanics, the pressure P is related to the kinetic energy K , the volume V , and the internal virial ϑ via the expression

$$P = \frac{1}{3V} (2K + \vartheta),$$

which can be used to calculate both the instantaneous pressure $P(t)$, as well as the average pressure $\langle P \rangle$ ($= \langle P(t) \rangle$) in numerical simulations. The angular brackets here denote time-averaging.

- a) The equation of state $P(\rho)$ of an interacting system, where P and $\rho = N/V$ are the pressure and the constant homogeneous number density, respectively, can be written as a virial expansion in order of ρ according to

$$\beta P/\rho = 1 + B_2\rho + B_3\rho^2, \quad (1)$$

where $\beta = \frac{1}{k_B T}$, and the second virial coefficient B_2 is defined through the pair potential $U(r)$ via

$$B_2 = -\frac{1}{2} \int d^3r [\exp(-\beta U(r)) - 1]. \quad (2)$$

Tasks: (4 points)

- i) Calculate numerically the second virial coefficient, B_2 of the Lennard-Jones interaction as a function of the LJ interaction energy, ϵ .
 - ii) Plot and interpret the function B_2/σ^3 vs $\epsilon/k_B T$ in the range $[0, 1]$. (To ensure adequate resolution, at least 15 points should be used, distributed uniformly over the interval $[0, 1]$.)
- b) Enrich the 3D molecular dynamics implementation from Exercise sheet #5 with a function that evaluates the pressure, P , of the system.

Using a time step $\Delta t = 2$ fs, simulate the trajectory of a system containing 8^3 atoms in a simulation box of volume V , such that the number density is given by $\rho = \frac{N}{V} = 0.05 \text{ } \sigma^{-3}$. The total simulation time should be 300 ps, divided into:

- 100 ps for the equilibration phase,
- 200 ps for the production run.

The temperature of the system should be equilibrated and maintained at $T_d = 300$ K using one of the thermostat schemes implemented in Exercise sheet #5 (e.g., Berendsen or Andersen thermostat).

Tasks: (8 points)

- i) Plot and report $P(t)$ and $T(t)$ as functions of time t during the simulation.
- ii) Compute and plot the radial distribution function $g(r)$ versus r , obtained from the production run as already done in sheet #3.
- iii) Divide the production run into 5 blocks of equal length and compute the average pressure $\langle P \rangle$ for each block.
- iv) Estimate and report the statistical error on $\langle P \rangle$.
- v) Validate your implementation by comparing the mean pressure $\langle P \rangle$ obtained from the simulation with the pressure predicted using the second virial coefficient.

Hint: For this exercise, you may use either the *Berendsen* or *Andersen* thermostat to control the temperature of the system.

- For the **Berendsen thermostat**, use a coupling time constant of $\tau = 50\Delta t$, where Δt is the integration time step.
- For the **Andersen thermostat**, use a collision frequency of $\nu = 0.01 \text{ fs}^{-1}$.

- c) Repeat the calculations and analysis as described in part (a), but now for a range of number densities given by $0.005 \sigma^{-3} \leq \rho \leq 0.25 \sigma^{-3}$.

The ideal gas pressure at a given temperature T_d is given by:

$$P_{\text{id}} = \rho k_B T_d.$$

Tasks: (8 points)

- i) For each value of ρ , compute the pressure P from the simulation and evaluate the deviation $P - P_{\text{id}}$.
- ii) Plot $P - P_{\text{id}}$ as a function of ρ . Investigate whether this deviation scales with ρ^2 .
- iii) Comment on your observations and provide a physical interpretation for the observed behavior. In particular, discuss how the nature of intermolecular interactions (repulsive vs. attractive) at different densities affects the deviation from ideal gas behavior.
- iv) Based on your results, predict the behavior of $P - P_{\text{id}}$ if the calculations were repeated with $\varepsilon = 0.1 k_B T$. Justify your prediction using insights from the virial expansion and the role of the second virial coefficient.

(Bonus (20 points)) Exercise 6.2: Isothermal-Isobaric (NPT) ensemble. The Berendsen barostat.

In computer simulations, a barostat is a scheme for controlling the pressure. Together with a thermostat, this can be used to perform simulations in the isothermal-isobaric (NPT) ensemble. In the following, a simple version of the Berendsen barostat is described, which uses isotropic expansion/compression of the system to control the pressure. All particle coordinates ($\mathbf{r}_i$), as well as the box dimensions (L_x, L_y, L_z) are scaled by the (same) scaling factor μ to achieve the isotropic expansion/compression.

If P_d is the desired pressure, or the *set* pressure, and $P(t)$ is the instantaneous pressure (calculated as described in Exercise 6.1), the scaling factor μ can be defined as

$$\mu(t) = \left[1 + \frac{\Delta t \kappa_T}{\tau_p} (P(t) - P_d) \right]^{\frac{1}{3}}$$

where Δt is the simulation time-step, κ_T is the isothermal compressibility of the system and τ_p is the coupling time constant for the pressure. Following the calculation of $\mu(t)$, the particle positions and dimensions are scaled as

$$\begin{aligned} \mathbf{r}_i(t) &\rightarrow \mu(t) \mathbf{r}_i(t) \\ L_k(t) &\rightarrow \mu(t) L_k(t); k \in \{x, y, z\} \end{aligned}$$

Note that κ_T is generally not *a-priori* known. However, the inaccuracy in κ_T in principle only changes the coupling time for pressure through the time constant τ_p (Berendsen *et al.*, J. Chem. Phys. **81** (8), 3684 (1984)). In other words, κ_T and τ_p together dictate the equilibration time for pressure.

- a) Implement the above algorithm in your simulation routine, such that it is used only when pressure control is desired. To validate your implementation, perform the following simulation. First, using a Berendsen thermostat with $\tau = 50 \Delta t$, equilibrate the temperature

of a system of $N = 8^3$ LJ particles with density $\rho = 0.5\sigma^{-3}$ at $T_d = 300K$ during 5000 time steps. At the end of the temperature equilibration, switch-on the Berendsen barostat with $\tau_p = 300\Delta t$, and $\kappa_T \sim 0.001 \text{ bar}^{-1}$, and equilibrate the pressure at $P_d = 50 \text{ bar}$ while maintaining the temperature at $T_d = 300K$ in the NPT ensemble during 50000 time steps. Finally, without changing any parameters, perform a production run of 50000 time steps. Plot the time evolutions of the instantaneous pressure $P(t)$ and instantaneous volume $V(t)$ for the entire duration of your simulation. What is the relaxation time of $P(t)$ during the equilibration run, and what are $\langle P \rangle$ and $\langle \sigma_P^2 \rangle$ during the production run (here $\langle P \rangle$ represents the time average of the pressure)?

- b) Repeat the simulation protocol as in (a), but now with $\tau_p \in (30\Delta t, 300\Delta t, 600\Delta t)$. For each value of τ_p , calculate and report the relaxation time of $P(t)$ during the equilibration run, and $\langle V \rangle$ and $\langle \sigma_V^2 \rangle$ during the production run. Comment on your observations.
- c) Repeat the simulation protocol as in (a), but now with $1 \text{ bar} < P_d < 1000 \text{ bar}$. Decompose the production run in 5 blocks of 10000 time steps, and calculate the mean pressure, $\langle P \rangle$, and the mean density, $\langle \rho \rangle = N / \langle V \rangle$, for each block. Calculate and report the corresponding statistical error for $\langle P \rangle$ and $\langle \rho \rangle$, and plot $\langle P \rangle$ as a function of $\langle \rho \rangle$. How does this equation of state compares with the one calculated in 6.1.c).

Numerical values of quantities to be used:

quantity	value (units)
k_B	0.0019849421 (kcal/mol/K)
ϵ	0.297741315 (kcal/mol)
σ	0.188 (nm)
T_d	300 (K)
mass	39.95 (g/mol)
Δt	2 (fs)
τ (Berendsen thermostat)	100 (fs)
ν (Andersen thermostat)	0.01 (fs ⁻¹)
κ_T (Berendsen barostat)	0.001 (bar ⁻¹)
τ_p (Berendsen barostat)	600 (fs)
P_d	50 (bar)